EXERCISE 3

The Wronskian of a Linear System: Liouville's Formula and the Dichotomy

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Wronskian • Liouville • Dichotomy

STATEMENT

Show that given a homogeneous linear system (1) with continuous coefficients in $[a, b]$, then the Wronskian $W(t)$ of two solutions (2) satisfies the first order ODE

$$\frac{dW}{dt} = (a_1(t) + b_2(t))\, W.$$

Deduce that this Wronskian is either identically 0 or nowhere zero in $[a, b]$.

SETUP

System (1) of Exercise 2 with a_1, b_1, a_2, b_2 continuous on $[a, b]$. For two solutions (x_1, y_1) and (x_2, y_2) the Wronskian is the determinant of the matrix whose columns are the solutions (Topic 4 Sec. 3):

$$W(t) = \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} = x_1 y_2 - x_2 y_1$$

FORMULATION

Differentiate W with the product rule, substitute the equations (H) of Exercise 2 for each derivative, and recognise W. The resulting scalar linear ODE is solved by an integrating factor.

CLASS NOTES AND COMPLEMENT [NOTES, P. 9]

The notes state that $\mathbf{x}_1, \mathbf{x}_2$ are independent if and only if $W = \det(x_{ij}) \neq 0$ for all $t \in I$; the words for all t are what this exercise proves. The equation $W' = (\operatorname{tr} A)\, W$ and its solution $W(t) = W(t_0) e^{\int_{t_0}^{t} \operatorname{tr} A}$ (Liouville's formula) are a complement (not in the class notes). For the system of a second-order equation $\operatorname{tr} A = -p$, and it becomes the $W' = -pW$ of Topic 3 (Exercise 6).

DERIVATION

Step 1: the product rule

$$\dot{W} = \dot{x}_1 y_2 + x_1 \dot{y}_2 - \dot{x}_2 y_1 - x_2 \dot{y}_1$$

Step 2: substitute the system

$$\dot{W} = (a_1 x_1 + b_1 y_1) y_2 + x_1 (a_2 x_2 + b_2 y_2) - (a_1 x_2 + b_1 y_2) y_1 - x_2 (a_2 x_1 + b_2 y_1)$$

Expand and collect by coefficient:

$$\dot{W} = a_1 (x_1 y_2 - x_2 y_1) + b_2 (x_1 y_2 - x_2 y_1) + b_1 (y_1 y_2 - y_2 y_1) + a_2 (x_1 x_2 - x_2 x_1)$$

The terms in b_1 and a_2 vanish identically ($y_1 y_2 - y_2 y_1 = 0$ and $x_1 x_2 - x_2 x_1 = 0$), and the remaining brackets are W:

$$\boxed{\dot{W} = (a_1 + b_2)\, W} \qquad (= (\operatorname{tr} A)\, W)$$

Step 3: solve the scalar equation

Let $\tau(t) = \int_{t_0}^{t} [a_1(s) + b_2(s)]\, ds$, which exists and is differentiable because $a_1 + b_2$ is continuous, with $\tau'(t) = a_1(t) + b_2(t)$ (fundamental theorem of calculus). The chain rule gives $\frac{d}{dt} e^{-\tau(t)} = -\tau'(t) e^{-\tau(t)}$, so by the product rule $\frac{d}{dt}\left(e^{-\tau} W\right) = e^{-\tau} \dot{W} - \tau' e^{-\tau} W = e^{-\tau} (\dot{W} - \tau' W) = e^{-\tau} \left(\dot{W} - (a_1 + b_2) W\right) = 0$, so $e^{-\tau} W$ is constant on $[a, b]$. Barrow's rule, $\left[e^{-\tau(s)} W(s)\right]_{t_0}^{t} = 0$, gives $e^{-\tau(t)} W(t) = e^{-\tau(t_0)} W(t_0) = W(t_0)$ (as $\tau(t_0) = 0$):

$$W(t) = W(t_0)\exp\left(\int_{t_0}^{t} [a_1(s) + b_2(s)]\,ds\right), \qquad t \in [a, b]$$

Step 4: the dichotomy

The exponential is finite and strictly positive, so $W(t) = 0$ exactly when $W(t_0) = 0$. Since $t_0 \in [a, b]$ was arbitrary: either $W(t_0) = 0$ for one (hence every) t_0, and then $W \equiv 0$ on $[a, b]$; or $W(t_0) \neq 0$ for one (hence every) t_0, and then W does not vanish anywhere on $[a, b]$. There is no third possibility. ■

INTERPRETATION AND CHECKS

- Constant coefficients: $W(t) = W(0)e^{(a_1+b_2)t} = W(0)e^{(\operatorname{tr} A)t}$ (Topic 4 Sec. 3, Liouville). Check with Example 28.1: eigenvalues 2 and -3 have trace -1 and $W \propto e^{2t}e^{-3t} = e^{-t}$.
- For the system of a second-order equation $\ddot{x} + p\dot{x} + qx = 0$, $a_1 = 0$ and $b_2 = -p$ and the result is Abel's identity $\dot{W} = -pW$ (Exercise 6).
- Geometric meaning: $|W|$ is the area of the parallelogram spanned by the two solution vectors; $\operatorname{tr} A$ is the divergence of the flow, which rates the growth of areas. If $\operatorname{tr} A = 0$ areas are conserved (W constant).
- This dichotomy is what makes the Wronskian a test for independence: it is enough to evaluate it at a single instant (Exercise 4).